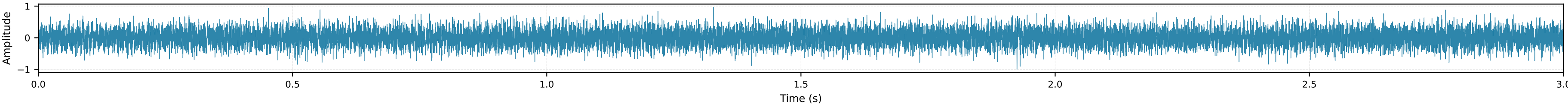


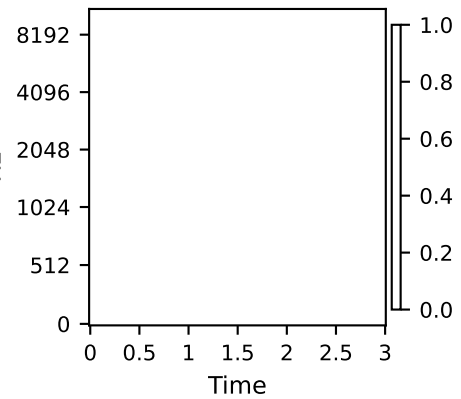
Preprocessing Pipeline: Raw Audio → 3-Channel Model Input



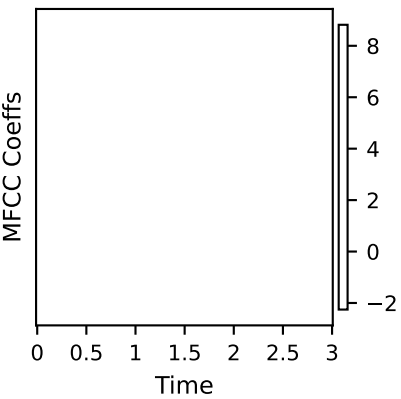
Step 1: Audio Waveform [SR=22050Hz, Duration=3.0s, Samples=66150]



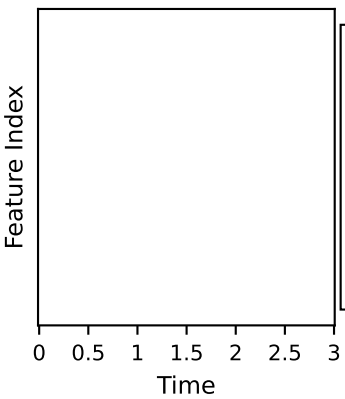
Step 2: Mel Spectrogram
Shape: (128, 130)



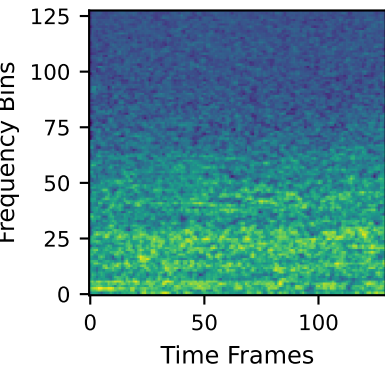
Step 3: MFCC
Shape: (120, 130)



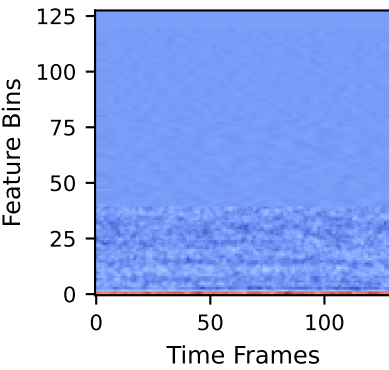
Step 4: Spectral Features
Shape: (9, 130)



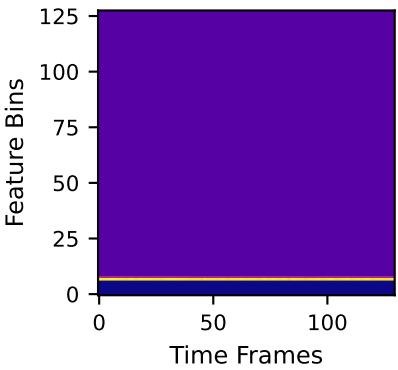
Channel 0: Mel
(128, 130)



Channel 1: MFCC
(128, 130)



Channel 2: Spectral
(128, 130)



Final 3-Channel Stack → Model Input: (3, 128, 130)

